

Supplementary Information

Different Alloy, Different Story? Impact of Transfer Learning on Predictive Models for In Silico Screening of Mg Dissolution Modulators

1 Data and sample counts

1.1 Per-alloy record counts

Table S1 lists the number of inhibition-efficiency (IE) records in the ExCorr database broken down by alloy. Alloys with fewer than 10 records are grouped as “misc.” for clarity. The three target alloys (AZ31, AZ91, WE43) and MgCa (contrast alloy, see main text Sec. 3.5) are highlighted.

Table S1: Per-alloy sample counts in the ExCorr Mg-inhibitor database. Record counts and unique canonical SMILES are shown for alloys with ≥ 10 records. “Role” indicates how the alloy is used in the experiments: target (AZ31/AZ91/WE43), close-transfer (all other Mg), or far-transfer (non-Mg base). MgCa is part of the close pool and is *additionally* used as the target alloy in the outside-cluster contrast experiment (Sec. 5).

Alloy	Base	Records	Unique mols	Role
AZ31	Mg	132	104	target
AZ91	Mg	121	100	target
WE43	Mg	120	104	target
AM50	Mg	122	104	close pool
ZE41	Mg	123	105	close pool
E21	Mg	117	102	close pool
CPMg220ppmFe	Mg	162	140	close pool
CPMg342ppmFe	Mg	70	62	close pool
HPMg50ppmFe	Mg	97	88	close pool
HPMg51ppmFe	Mg	87	80	close pool
<i>MgCa</i>	Mg	56	49	<i>close pool / contrast</i>
misc. Mg (<10 records)	Mg	5	–	close pool
Carbon Steel	Fe	1285	411	far pool
powder	Al	542	454	far pool
pure	Al	464	74	far pool
unknown pure	Al	459	70	far pool
AA2024-T3	Al	324	81	far pool
AA7075-T6	Al	292	80	far pool
AA1100	Al	144	6	far pool
AA1060	Al	122	14	far pool
other (<100 records)	–	~320	–	far pool
Total Mg		373 target + 839 close pool (incl. 56 MgCa) = 1,212		
Total far pool (non-Mg)		3,955 records, 1,097 molecules		

1.2 Per-fold test/pool molecule overlap

Table S2 reports, for each (alloy, fold) combination, how many of the 15 test-fold molecules also appear in the close and far transfer pools. On average, all 15 test molecules recur in the close

pool, each appearing on 5–8 other Mg materials. The *removal count* column gives the number of source-pool records eliminated by the molecule-disjoint filter used throughout the main experiments.

Table S2: Per-fold overlap between test-fold molecules and transfer pools. “in close” = number of test molecules (out of 15) whose canonical SMILES appears at least once in the close Mg pool; “removed” = number of close-pool records eliminated by the molecule-disjoint filter. Analogous columns for the far pool.

Alloy	Fold	in close	removed	in far	removed
AZ31	0	15	84	4	25
AZ31	1	15	104	4	50
AZ31	2	15	87	7	31
AZ31	3	15	110	4	35
AZ31	4	15	99	7	35
AZ91	0	15	82	5	26
AZ91	1	15	108	4	50
AZ91	2	15	87	6	35
AZ91	3	15	100	6	41
AZ91	4	15	117	7	53
WE43	0	15	85	4	23
WE43	1	15	106	3	49
WE43	2	15	85	7	16
WE43	3	15	104	4	30
WE43	4	15	89	10	55
mean		15.0	96.5	5.5	36.9

2 Full per-model per-setting metrics

2.1 Coefficient of determination R^2

Table S3 gives the mean $\pm$ standard error of R^2 between predicted and experimental IE, over the 15 (target alloy, fold) combinations of the main experiment. R^2 is the primary metric used in the main text.

Table S3: Mean test-set $R^2 \pm$ SE for each (model, setting) cell, averaged over 3 target alloys $\times$ 5 folds. Negative R^2 values indicate predictors worse than always predicting the pool mean IE. The Gemini 3.1 Pro row (averaged over 3 in-context calls per fold rather than the 15–25 seeds of the fitted models) is listed separately for reference, here and in Tables S4–S5.

	Exact (60)	Close filt.	Close unfilt.	Far filt.	Far unfilt.
RF	-0.11 ± 0.08	-0.14 ± 0.10	$+0.44 \pm 0.05$	-0.12 ± 0.09	$+0.40 \pm 0.05$
GBR	-0.14 ± 0.11	-0.16 ± 0.11	$+0.68 \pm 0.04$	-0.08 ± 0.07	$+0.58 \pm 0.04$
MLP	-0.10 ± 0.08	-0.11 ± 0.06	$+0.50 \pm 0.11$	-0.17 ± 0.07	$+0.54 \pm 0.10$
kNN-Tan	-0.19 ± 0.09	-1.19 ± 0.56	-0.14 ± 0.28	-0.49 ± 0.17	-0.10 ± 0.26
ChemProp	-0.14 ± 0.06	-0.49 ± 0.27	$+0.48 \pm 0.04$	-0.15 ± 0.11	$+0.42 \pm 0.06$
mean (fitted)	-0.13	-0.42	+0.39	-0.20	+0.37
Gemini 3.1 Pro	$+0.04 \pm 0.20$	$+0.11 \pm 0.21$	$+0.36 \pm 0.11$	$+0.19 \pm 0.12$	$+0.47 \pm 0.12$

2.2 Pearson correlation r

Table S4 reports the corresponding Pearson r values. r is included for comparison with prior literature and to make the calibration-vs-ranking gap visible: r is consistently high in the unfiltered settings, while R^2 (Table S3) exposes the bias that r cannot detect.

Table S4: Mean test-set Pearson $r \pm$ SE.

	Exact (60)	Close filt.	Close unfilt.	Far filt.	Far unfilt.
RF	$+0.27 \pm 0.07$	$+0.30 \pm 0.07$	$+0.81 \pm 0.03$	$+0.30 \pm 0.07$	$+0.79 \pm 0.04$
GBR	$+0.28 \pm 0.06$	$+0.30 \pm 0.08$	$+0.88 \pm 0.02$	$+0.26 \pm 0.07$	$+0.85 \pm 0.02$
MLP	$+0.29 \pm 0.07$	$+0.22 \pm 0.06$	$+0.84 \pm 0.03$	$+0.23 \pm 0.05$	$+0.84 \pm 0.03$
kNN-Tan	$+0.24 \pm 0.07$	$+0.19 \pm 0.08$	$+0.87 \pm 0.02$	$+0.23 \pm 0.09$	$+0.83 \pm 0.03$
ChemProp	$+0.18 \pm 0.08$	$+0.29 \pm 0.07$	$+0.75 \pm 0.03$	$+0.32 \pm 0.08$	$+0.70 \pm 0.04$
mean	+0.25	+0.26	+0.83	+0.27	+0.80
Gemini 3.1 Pro	$+0.63 \pm 0.05$	$+0.68 \pm 0.04$	$+0.79 \pm 0.04$	$+0.66 \pm 0.04$	$+0.81 \pm 0.03$

2.3 Mean absolute error (MAE)

Table S5 gives the test-set MAE in percentage-point units of IE.

Table S5: Mean test-set MAE (% IE) $\pm$ SE.

	Exact (60)	Close filt.	Close unfilt.	Far filt.	Far unfilt.
RF	107 $\pm$ 9	107 $\pm$ 9	75 $\pm$ 6	106 $\pm$ 9	78 $\pm$ 7
GBR	111 $\pm$ 8	110 $\pm$ 9	56 $\pm$ 5	109 $\pm$ 9	71 $\pm$ 7
MLP	124 $\pm$ 8	124 $\pm$ 9	56 $\pm$ 5	111 $\pm$ 9	58 $\pm$ 5
kNN-Tan	114 $\pm$ 7	127 $\pm$ 8	79 $\pm$ 5	114 $\pm$ 10	78 $\pm$ 5
ChemProp	128 $\pm$ 10	110 $\pm$ 6	79 $\pm$ 6	108 $\pm$ 6	82 $\pm$ 7
Gemini 3.1 Pro	94 $\pm$ 8	88 $\pm$ 8	75 $\pm$ 5	90 $\pm$ 6	68 $\pm$ 4

3 Cross-alloy analysis

Figure S1 breaks out the pooled R^2 of main text Fig. 1 into the three target alloys separately. The filter-sensitivity pattern—peak at Close-unfiltered, collapse to the Exact baseline once the molecule-disjoint filter is applied—is present on every individual alloy, ruling out that the pooled mean averages over alloys that behave qualitatively differently.

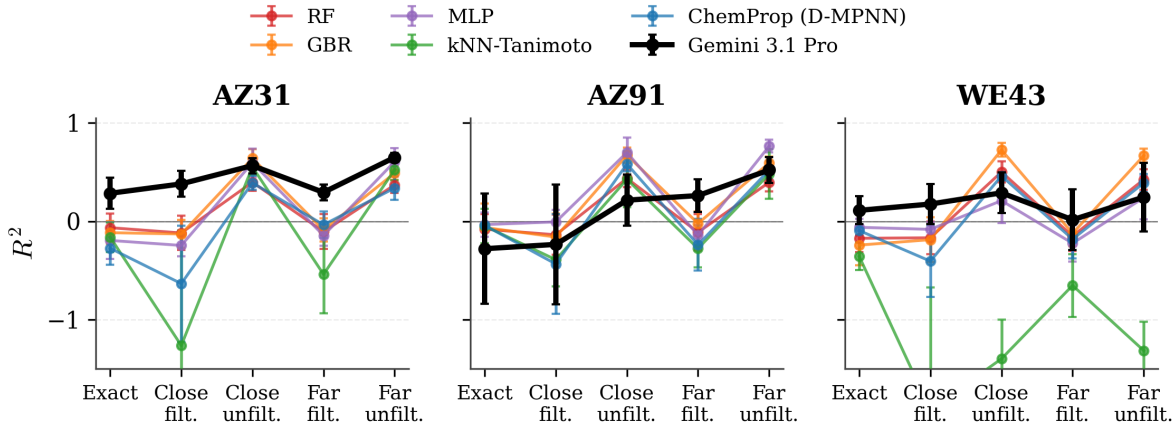


Figure S1: **Per-target-alloy R^2 across settings.** Same as the R^2 panel of main text Fig. 1, broken out into three panels (one per target alloy).

Quantitative differences between the three alloys are visible in Table S6. AZ31 and AZ91 yield the strongest Close-unfiltered fits (mean across the four calibrated fitted models $R^2 \approx +0.51$ and $+0.60$), while WE43 is weaker ($R^2 \approx +0.47$ excluding the k NN-Tanimoto outlier, which collapses to $R^2 = -1.40$ on WE43). The same ordering broadly tracks each alloy’s mean correlation with its cluster neighbours (Fig. 3 of the main text), consistent with the interpretation of Sec. 3.3: the closer a target alloy sits to the cluster centroid, the larger the alloy-context re-scaling term that dominates the apparent TL gain.

Gemini 3.1 Pro is one of the most alloy-stable methods in the Close-filtered and Exact settings and remains the only method retaining non-trivial predictive accuracy in the filtered and Exact settings on each alloy individually. Interpretation of the LLM result is given in Sec. 3.6 of the main text.

Table S6: Per-target-alloy mean $R^2 \pm \text{SE}$ at the Close-unfiltered setting (5-fold cross-validation).

Alloy	RF	GBR	MLP	kNN-Tan	ChemProp
AZ31	$+0.40 \pm 0.09$	$+0.64 \pm 0.09$	$+0.59 \pm 0.14$	$+0.55 \pm 0.08$	$+0.39 \pm 0.07$
AZ91	$+0.44 \pm 0.09$	$+0.67 \pm 0.08$	$+0.70 \pm 0.15$	$+0.43 \pm 0.28$	$+0.58 \pm 0.08$
WE43	$+0.50 \pm 0.11$	$+0.73 \pm 0.07$	$+0.21 \pm 0.23$	-1.40 ± 0.40	$+0.46 \pm 0.04$

4 Cluster robustness: bootstrap CIs and anchor controls

The main-text cluster claim (Fig. 3, seven Mg-based materials—six commercial alloys plus the high-purity variant HPMg51—at pairwise Spearman $r = 0.67$ – 0.92) rests on a rank-correlation matrix computed over per-molecule mean IE on the intersection of molecules measured on both materials in each pair. Two concerns about this construction are addressed here: (i) whether the pairwise r estimates are stable under resampling given the limited number of shared molecules per pair, and (ii) whether the cluster structure is driven by a small set of high-IE anchor compounds (e.g. urea, benzotriazole, picolinic acid) that recur across materials.

For each pair we resample the shared molecules with replacement 1000 times and recompute Spearman r , and we additionally recompute r' after excluding the top-decile-IE molecules of that pair (ranked by the pair’s mean transformed IE). Table S7 reports n (shared molecules), r with 95% bootstrap CI, r' and n' (molecules retained after anchor exclusion).

Commercial cluster is robust. Across the 21 unique pairs of these seven materials, shared overlaps are large (73–100 molecules, median 96) and every 95% CI lower bound is *positive* (range 0.52–0.87). The weakest cluster edge (E21–ZE41, $r = 0.67$) has CI [0.52, 0.77]. The cluster is therefore not explicable as sampling noise on a shared anchor set. The anchor-exclusion partial correlation r' drops modestly from a median of 0.80 to 0.74 (range 0.55–0.89): all pairs remain inside the cluster band, so the cluster structure is not carried by the extreme inhibitors alone—the ordering among moderate-IE molecules is itself strongly conserved across the seven cluster materials.

MgCa outlier status is confirmed, but individual values should not be over-interpreted.

The ten MgCa pairs have smaller shared-molecule counts (11–45, versus 73–100 within the commercial cluster), which translates into wider CIs, but every point estimate stays ≤ 0.35 and every CI spans or includes zero. The anchor-exclusion r' values span $[-0.26, +0.22]$ with no systematic direction, consistent with a near-zero underlying correlation and small n . No transformation of the underlying data produces a MgCa pair that would fall inside the commercial-cluster range. Because of the small overlaps, individual MgCa pairwise r values carry substantial sampling uncertainty and should not be read as precise quantities; the conclusion this analysis supports is the qualitative one — that MgCa lies uniformly outside the commercial-cluster band ($r \geq 0.67$) — not any specific numerical r .

Table S7: Pairwise Spearman robustness: n shared molecules, r with 95% bootstrap CI (1000 resamples), r' after excluding the top-decile-IE molecules of the pair, and n' molecules retained. Rows ordered by pair kind (commercial-cluster pairs, then MgCa-pairs, then other pairs).

Pair	Kind	n	r	95% CI	r'	n'
AM50–AZ31	commercial	98	0.91	[0.86, 0.94]	0.89	88
AM50–AZ91	commercial	94	0.92	[0.86, 0.94]	0.89	84
AM50–E21	commercial	98	0.78	[0.67, 0.86]	0.71	88
AM50–HPMg51	commercial	74	0.82	[0.73, 0.88]	0.76	66
AM50–WE43	commercial	99	0.83	[0.73, 0.90]	0.77	89
AM50–ZE41	commercial	99	0.77	[0.68, 0.85]	0.73	89
AZ31–AZ91	commercial	94	0.86	[0.77, 0.92]	0.80	84
AZ31–E21	commercial	95	0.75	[0.64, 0.83]	0.66	85
AZ31–HPMg51	commercial	73	0.82	[0.73, 0.88]	0.76	65
AZ31–WE43	commercial	97	0.80	[0.70, 0.87]	0.74	87
AZ31–ZE41	commercial	100	0.83	[0.74, 0.89]	0.77	90
AZ91–E21	commercial	96	0.77	[0.66, 0.84]	0.69	86
AZ91–HPMg51	commercial	73	0.78	[0.68, 0.86]	0.73	65
AZ91–WE43	commercial	97	0.80	[0.70, 0.87]	0.73	87
AZ91–ZE41	commercial	95	0.74	[0.63, 0.83]	0.65	85
E21–HPMg51	commercial	75	0.86	[0.75, 0.93]	0.82	67
E21–WE43	commercial	100	0.84	[0.74, 0.91]	0.79	90
E21–ZE41	commercial	97	0.67	[0.52, 0.77]	0.55	87
HPMg51–WE43	commercial	77	0.89	[0.82, 0.93]	0.85	69
HPMg51–ZE41	commercial	75	0.73	[0.59, 0.81]	0.63	67

(continued on next page)

Table S7 (continued)

Pair	Kind	n	r	95% CI	r'	n'
WE43–ZE41	commercial	99	0.79	[0.68, 0.87]	0.71	89
AM50–MgCa	MgCa	14	0.24	[-0.55, 0.77]	-0.20	12
AZ31–MgCa	MgCa	15	0.31	[-0.27, 0.73]	0.14	13
AZ91–MgCa	MgCa	11	0.29	[-0.54, 0.90]	-0.13	9
CPMg220–MgCa	MgCa	21	0.05	[-0.33, 0.47]	-0.15	18
CPMg342–MgCa	MgCa	45	0.08	[-0.24, 0.41]	0.07	40
E21–MgCa	MgCa	13	0.24	[-0.45, 0.72]	-0.26	11
HPMg50–MgCa	MgCa	15	0.25	[-0.27, 0.73]	0.22	13
HPMg51–MgCa	MgCa	13	0.21	[-0.47, 0.64]	-0.15	11
MgCa–WE43	MgCa	14	0.16	[-0.48, 0.62]	-0.21	12
MgCa–ZE41	MgCa	15	0.35	[-0.26, 0.79]	0.12	13
AM50–CPMg220	other	96	0.52	[0.33, 0.66]	0.41	86
AM50–CPMg342	other	20	0.27	[-0.22, 0.63]	0.03	18
AM50–HPMg50	other	83	0.42	[0.24, 0.59]	0.35	74
AZ31–CPMg220	other	95	0.63	[0.46, 0.75]	0.53	85
AZ31–CPMg342	other	22	0.44	[-0.01, 0.73]	0.20	19
AZ31–HPMg50	other	83	0.57	[0.40, 0.70]	0.48	74
AZ91–CPMg220	other	90	0.54	[0.36, 0.68]	0.43	81
AZ91–CPMg342	other	18	0.59	[0.03, 0.84]	0.42	16
AZ91–HPMg50	other	78	0.48	[0.29, 0.63]	0.39	70
CPMg220–CPMg342	other	36	0.55	[0.27, 0.75]	0.41	32
CPMg220–E21	other	93	0.45	[0.25, 0.61]	0.31	83
CPMg220–HPMg50	other	80	0.65	[0.48, 0.78]	0.59	72
CPMg220–HPMg51	other	70	0.55	[0.35, 0.71]	0.45	63
CPMg220–WE43	other	95	0.58	[0.43, 0.69]	0.48	85
CPMg220–ZE41	other	96	0.68	[0.54, 0.78]	0.61	86
CPMg342–E21	other	20	0.39	[-0.10, 0.76]	0.17	18
CPMg342–HPMg50	other	19	0.37	[-0.06, 0.69]	0.25	17
CPMg342–HPMg51	other	20	0.51	[0.06, 0.79]	0.36	18
CPMg342–WE43	other	21	0.40	[-0.10, 0.77]	0.12	18
CPMg342–ZE41	other	22	0.33	[-0.10, 0.62]	0.10	19
E21–HPMg50	other	81	0.33	[0.12, 0.52]	0.24	72
HPMg50–HPMg51	other	74	0.39	[0.15, 0.60]	0.26	66
HPMg50–WE43	other	84	0.47	[0.28, 0.61]	0.38	75
HPMg50–ZE41	other	85	0.61	[0.45, 0.74]	0.53	76

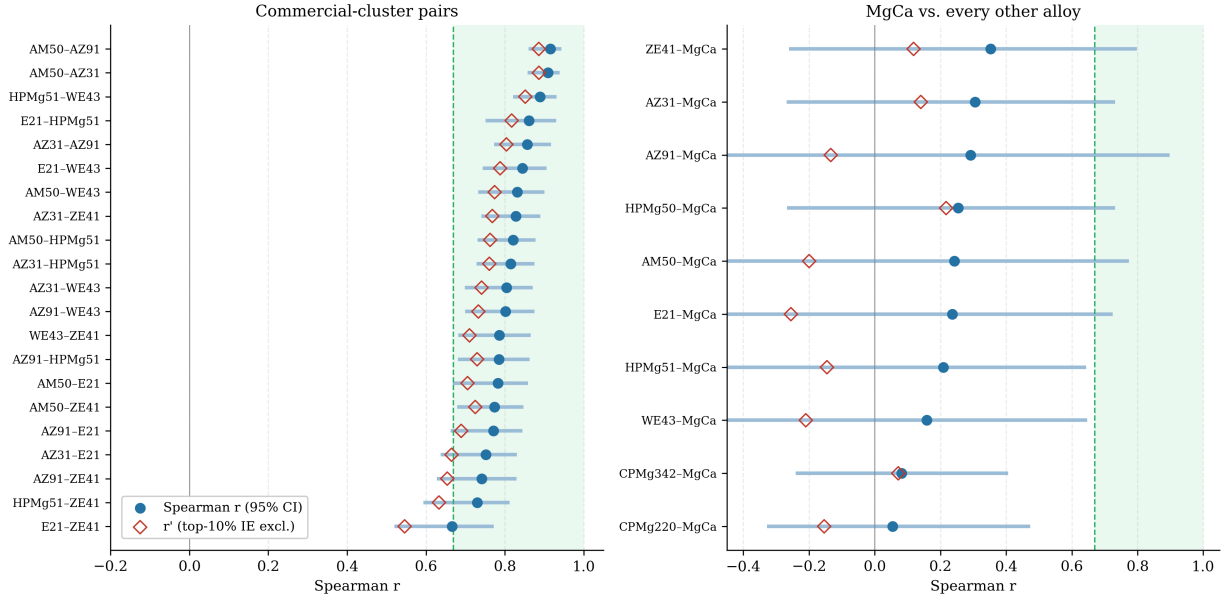


Figure S2: **Cluster robustness.** Horizontal forest plot of the pairwise Spearman r (blue circles) with 95% bootstrap CIs (horizontal bars) and the anchor-exclusion r' (red open diamonds), pairs sorted by r . The vertical dashed line and green shaded band mark the cluster range ($r \geq 0.67$). Left: the 21 unique commercial-cluster pairs—every CI lies above zero and r' tracks r closely. Right: MgCa against every other material—point estimates cluster near zero with CIs that mostly cross zero, and r' is similarly uninformative.

4.1 Pairwise IE scatter examples

To make the cluster geometry concrete we plot raw inhibitor-efficiency (IE) values directly against each other for representative material pairs. Each point is one canonical SMILES measured on both materials. Axes use a symmetric-log scale (linear within $\pm 20\%$ IE, logarithmic beyond) spanning -500 to $+100\%$ IE, so the strong-accelerator tail does not compress the inhibition region; the few points beyond this range are omitted from the plot only, not from the reported Spearman r .

Cluster regularity from one alloy’s perspective. Figure S3 plots AM50’s IE against each of the six other commercial-cluster members. The same near-diagonal trend appears in every panel ($r = 0.78\text{--}0.92$, $n = 74\text{--}99$): IE values on AM50 track IE values on the cluster neighbour by a consistent monotonic rescaling, including in the strong-accelerator region ($\text{IE} < 0$). The pattern is consistent regardless of which cluster member is paired with AM50, supporting the cluster interpretation as a property of the seven materials jointly rather than of any specific pair.

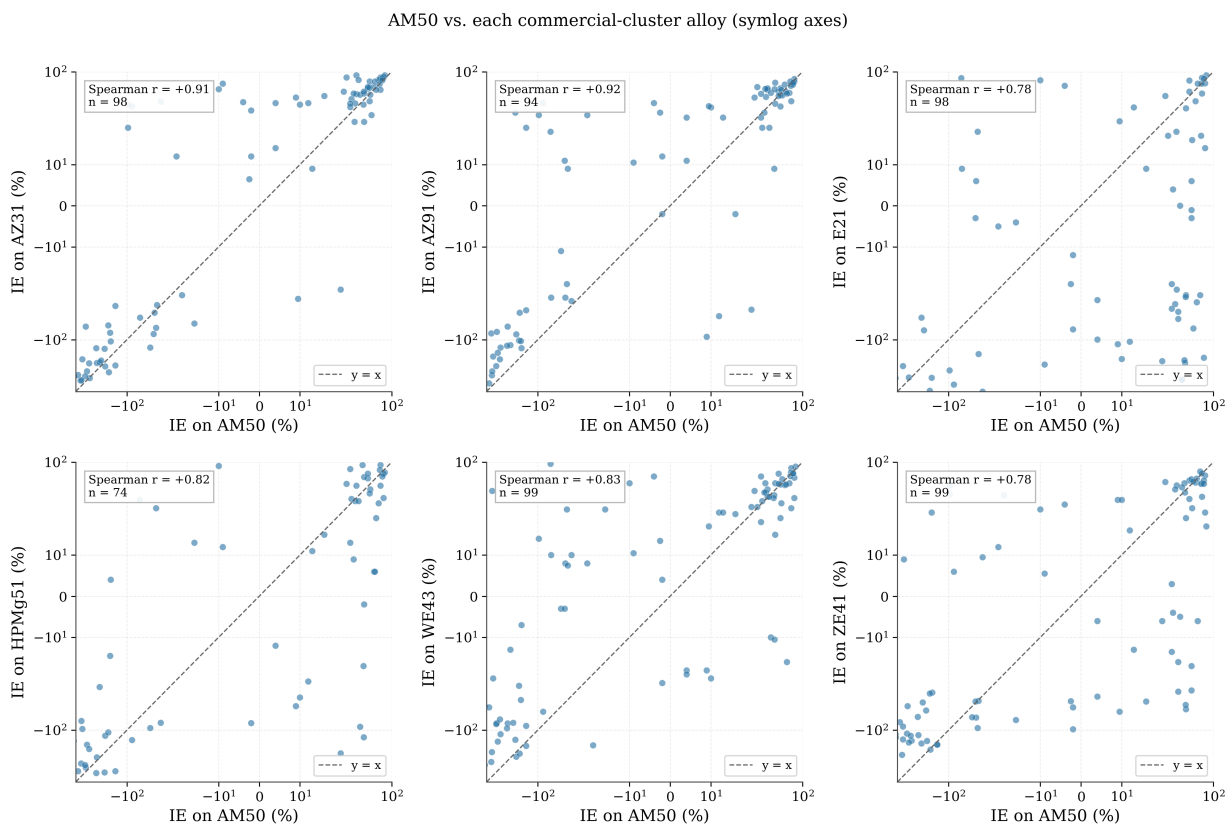


Figure S3: **AM50 IE versus each other commercial-cluster member.** Six panels showing AM50 (x-axis) against AZ31, AZ91, E21, HPMg51, WE43 and ZE41 on the y-axis, on symmetric-log axes (see text). The $y = x$ identity line is drawn in each panel. All six pairs display the same near-diagonal cluster pattern; the slight reduction in r for E21 and ZE41 ($r = 0.78$) corresponds to a few large-magnitude accelerator molecules that dominate the Spearman ranks at the low-IE end.

Outlier and high-Fe pairs. Figure S4 contrasts a cluster-outlier pair (ZE41 vs MgCa) with the high-Fe purity pair (HPMg50 vs HPMg51).

Representative alloy pairs in IE-space (symlog axes)

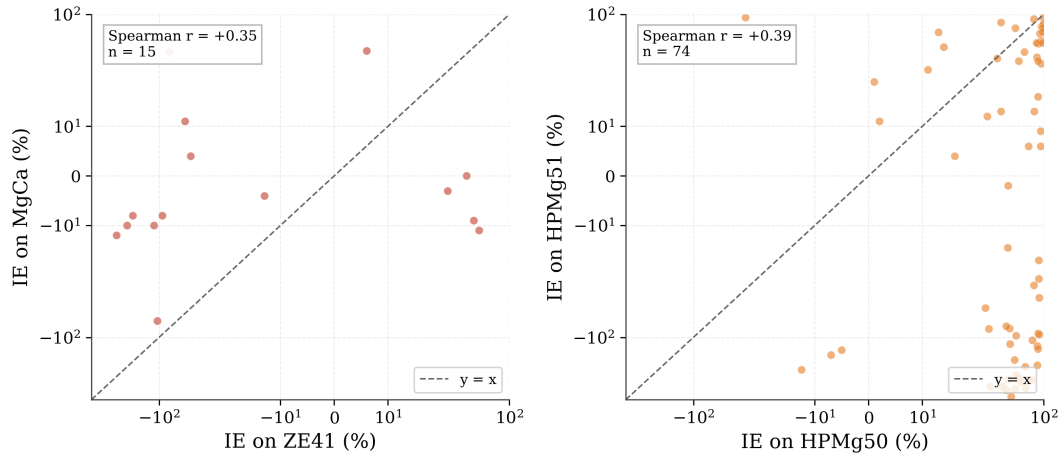


Figure S4: **Two atypical material pairs in IE-space** (symmetric-log axes, see text). Left: ZE41 vs MgCa, the cluster-outlier pair with the largest molecular overlap with MgCa ($n=15$, tied with AZ31-MgCa) — IE values on MgCa stay close to zero regardless of the ZE41 IE, with no diagonal pattern (Spearman $r = +0.35$). Right: HPMg50 vs HPMg51, the high-Fe purity pair — despite nominally similar compositions and Fe levels these two materials correlate substantially worse than any commercial-cluster pair (Spearman $r = +0.39$, $n = 74$).

5 MgCa contrast experiment: full results

Table S8 gives the complete MgCa contrast results. The k NN-Tanimoto baseline, which reaches $r = +0.87$ on target alloys, drops to $r = +0.14$ on MgCa at the close-unfiltered setting, confirming that when the source pool’s molecules do not share a mechanism with the target alloy, even direct similarity lookup fails. Gemini 3.1 Pro, run on MgCa with the same in-context protocol as in the main experiment (Table S9), retains a moderate ranking signal ($r = +0.46$ to $+0.64$) but—like the fitted models—does not reach positive R^2 on this outside-cluster alloy; crucially, it never collapses to the strongly negative R^2 that cross-alloy transfer inflicts on the fitted models, consistent with the main-text interpretation that its pretraining keeps predictions on a sensible IE scale (main text Sec. 3.5).

Table S8: MgCa contrast experiment: mean test-set Pearson $r \pm$ SE (4-fold CV, 56 total records). Compare with Table S4: in-cluster target alloys reach $r > +0.80$ at close-unfiltered, whereas the five *fitted* models never exceed $r = +0.19$ on MgCa under any setting. Gemini 3.1 Pro is the exception, reaching $r = +0.46$ to $+0.64$; its corresponding R^2 and MAE are given in Table S9.

	Exact	Close filt.	Close unfilt.	Far filt.	Far unfilt.
RF	-0.57 ± 0.13	-0.01 ± 0.18	-0.04 ± 0.11	-0.00 ± 0.19	-0.04 ± 0.10
GBR	-0.53 ± 0.13	$+0.16 \pm 0.12$	$+0.09 \pm 0.10$	-0.05 ± 0.17	-0.06 ± 0.06
MLP	-0.16 ± 0.02	$+0.01 \pm 0.09$	$+0.19 \pm 0.05$	-0.14 ± 0.17	$+0.02 \pm 0.10$
k NN-Tan	-0.03 ± 0.11	-0.15 ± 0.09	$+0.14 \pm 0.20$	-0.14 ± 0.07	-0.05 ± 0.08
ChemProp	-0.15 ± 0.13	$+0.00 \pm 0.11$	$+0.07 \pm 0.11$	$+0.08 \pm 0.18$	$+0.14 \pm 0.20$
Gemini 3.1 Pro	$+0.54 \pm 0.13$	$+0.52 \pm 0.14$	$+0.48 \pm 0.16$	$+0.64 \pm 0.09$	$+0.46 \pm 0.22$

Table S9: MgCa contrast, Gemini 3.1 Pro: mean test-set R^2 and MAE $\pm$ SE (4-fold CV, in-context, 3 calls/fold). Despite the moderate Pearson r above, R^2 stays near zero in every setting—Gemini does not achieve positive variance explained on this outside-cluster alloy, but it also avoids the catastrophic miscalibration of the fitted models (whose MgCa R^2 falls to -3 to < -50 ; main text Sec. 3.5). MAE is low only because MgCa IE values are themselves small.

	Exact	Close filt.	Close unfilt.	Far filt.	Far unfilt.
R^2	-0.22 ± 0.46	-0.29 ± 0.46	$+0.02 \pm 0.30$	-0.20 ± 0.45	$+0.11 \pm 0.23$
MAE (% IE)	11.3 ± 2.5	11.2 ± 1.8	10.5 ± 1.6	11.2 ± 2.5	10.2 ± 2.0

6 Feature and model details

6.1 Feature construction

All fingerprint-based models receive the same input block:

- Morgan fingerprint (2,048 bits, radius 2, chirality-aware)
- Context features: HCl %, NaCl %, operating concentration [mM]
- Alloy one-hot encoding for *Close* and *Far* settings, spanning all 40 distinct alloy labels in the ExCorr database
- Base-material one-hot encoding for *Far* settings (Mg, Al, Fe, Cu, Zn)

In the *Exact* setting, no alloy one-hot is provided because the training set is a single alloy—the column would be constant. The effective input dimensions per setting are:

Table S10: Feature-vector dimension by transfer setting.

Setting	Dim. (FP + OHE + context)
Exact	2048 + 3 context = 2051
Close (filt./unfilt.)	2051 + 40 alloy OHE = 2091
Far (filt./unfilt.)	2091 + 5 base OHE = 2096

ChemProp uses the same context/OHE block as extra descriptors (“**x.d**”) concatenated to the graph embedding before the feed-forward regression head.

k NN-Tanimoto uses only the Morgan fingerprint (Tanimoto distance cannot meaningfully combine fingerprint bits with continuous context features); it is therefore slightly disadvantaged in Exact and Close settings but unaffected by Far information it cannot use.

6.2 Hyperparameter summary

Table S11: Model hyperparameters. All models are averaged over 25 random seeds (15 for ChemProp).

Model	Hyperparameters
k NN-Tanimoto	$k = 5$, distance-weighted mean, Tanimoto
Random Forest	200 trees, scikit-learn defaults
Gradient Boosting	200 trees, depth 5, lr 0.05, subsample 0.8
MLP	256-128-64 hidden, ReLU, early stopping on 15% val
ChemProp D-MPNN	hidden 300, 5 FFN layers, 5 epochs, batch 8, lr 10^{-5}

6.3 In-context LLM prompt

The Gemini 3.1 Pro predictions follow the in-context protocol of our earlier study on LLM-based Mg-inhibitor prediction (cited in the main text) — the “**with_prealysis**”, names-only configuration. Each query is assembled from (i) a fixed chemist-role system prompt, (ii) the per-fold training records rendered as labelled examples, and (iii) the held-out molecule with its experimental conditions but no label. The model is asked for a single inhibition-efficiency value and the prediction is averaged over 3 independent calls. The verbatim prompt strings are included in the released code.

System prompt (chemist role, abridged).

You are a professional chemist with deep knowledge about organic chemistry and corrosion mechanisms. [...] You are tasked with predicting the inhibition efficiency of an organic compound from its molecular structure and the experimental conditions. The inhibition efficiency is expressed in percent (at most 100); negative values denote acceleration of corrosion. It is a *relative* value with respect to an inhibitor-free control under otherwise identical conditions, so the reference differs for each alloy and environment.

In-context training examples. The (optionally molecule-disjoint-filtered) per-fold training records are concatenated, each rendered as:

Here is additional data from previous experiments that might help you:

Question: Predict the inhibition efficiency of the following setting:

IUPAC: <name>

SMILES: <SMILES>

Corrosion sample: <base material>, <alloy>

Concentration inhibitor mM: <concentration>

Aggressive environmental component: <electrolyte>

Answer: Predicted inhibition efficiency: <IE>%.

Query. The held-out molecule is presented in the same field format but *without* the **Answer** line; the model returns one predicted inhibition efficiency for it.

6.4 Compute

The main experiment ($3 \text{ target alloys} \times 5 \text{ folds} \times 5 \text{ settings} \times 5 \text{ models} = 375 \text{ combinations}$) and the MgCa contrast ($4 \text{ folds} \times 5 \text{ settings} \times 5 \text{ models} = 100 \text{ combinations}$) were run on the TUHH HPC cluster, one SLURM job per combination with 2 CPU cores and 8 GB of RAM. Classical model combinations (kNN/RF/GBR/MLP) complete in 5–60 minutes each; ChemProp combinations take 20–180 minutes depending on setting. Total wall time ≈ 20 core-hours per experiment.

7 Paired-test results

Table S12 reports the paired Wilcoxon signed-rank test comparing each model’s Exact-setting correlation to its Close-filtered correlation, over the 15 folds. Across all five models, we cannot reject the null hypothesis that the two settings yield the same test-set correlation: the molecule-disjoint filter leaves no detectable cross-alloy TL signal.

Table S12: Paired Wilcoxon signed-rank test: Exact vs Close-filtered. Null hypothesis: the two settings have equal median Pearson r .

Model	Wilcoxon p	Reject H_0 ?
RF	0.52	No
GBR	0.80	No
MLP	0.42	No
kNN-Tan	0.64	No
ChemProp	0.15	No
pooled	0.62	No

8 Gemini 3.1 Pro: zero-shot contamination check

Table S13 reports the per-alloy Pearson r , R^2 and MAE for the zero-shot Gemini 3.1 Pro variant described in Sec. 3.6 of the main text, in which the model receives the same chemist-role system prompt and experimental-condition fields as in the in-context setting but *no* in-context training rows. Each of the 75 molecules per target alloy is queried 3 times. Interpretation of these numbers—including the comparison with in-context close-filtered $r = +0.68$ and the conclusion that the in-context performance is not driven by pretraining-corpus recall of ExCorr labels—is given in the main text.

Table S13: Zero-shot Gemini 3.1 Pro on the three target alloys (no in-context training rows). Pearson r near zero, negative R^2 , and MAE roughly $1.5\times$ that of the in-context filtered setting collectively rule out label-level memorisation. p -values are two-sided Pearson tests on the per-alloy ($n=75$) and pooled ($n=225$) molecule-level predictions.

Alloy	Pearson r	p	R^2	MAE (% IE)
AZ31	+0.16	0.16	−0.30	147
AZ91	+0.22	0.061	−0.17	142
WE43	+0.15	0.20	−0.66	101
pooled ($n=225$)	+0.16	0.016	−0.28	130

The per-alloy zero-shot Pearson tests fail to reach $\alpha=0.05$ on every target alloy ($p=0.06$ – 0.20). Only when the 225 molecule-level predictions are pooled across the three target alloys does the test reach $p=0.016$, and the correspondingly weak $r = +0.16$ is an order of magnitude below the in-context Gemini Pearson r at any setting ($r = +0.59$ to $+0.80$ on the same pooled-prediction test, all with two-sided $p < 10^{-22}$; Table S14). Zero-shot Gemini therefore carries at most a marginal qualitative prior over Mg-inhibitor IE ranking, well short of what would be expected if pretraining had memorised the ExCorr (molecule, IE) labels.

9 Pearson correlation and mean absolute error

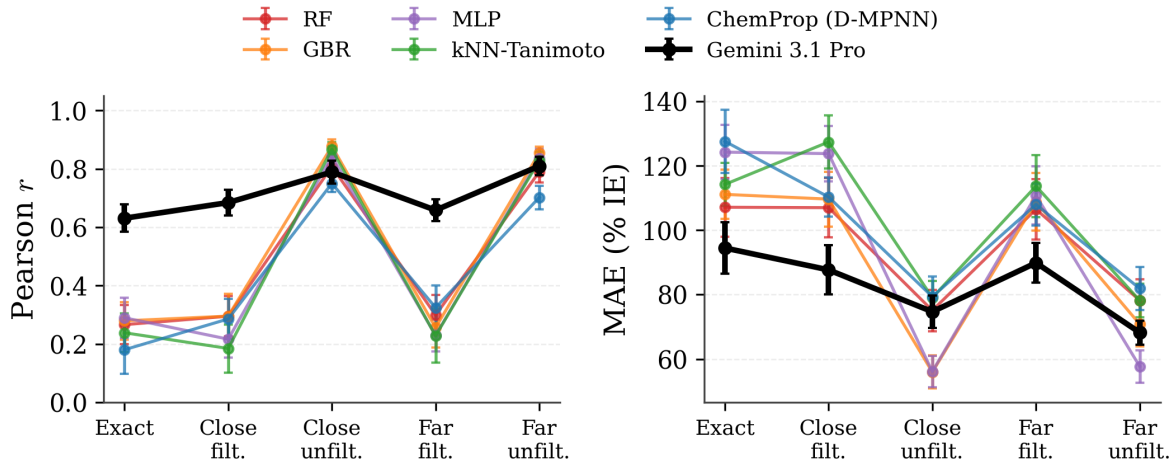


Figure S5: **Pearson r and mean absolute error companion to main text Fig. 1.** Mean over the 15 (target alloy, fold) combinations. Both Pearson r and MAE reproduce the filter-sensitivity pattern of R^2 : close-unfiltered TL reaches $r > 0.8$ and MAE below 80 % IE for the best fitted models, while filtered and Exact settings drop r to $\approx +0.25$ with MAE above 100 % IE. Pearson r is more permissive than R^2 on k NN-Tanimoto because the lookup ranks molecules well but does not calibrate absolute IE (close-unfiltered $r = +0.87$ but $R^2 = -0.14$).

Pearson r and MAE confirm the R^2 conclusions of main text Sec. 3.1–3.2 from a ranking-only and an absolute-error perspective respectively: the close-unfiltered gain is real in both metrics; the filtered collapse extends to all three; and the in-context LLM retains its filtered-setting advantage over every fitted baseline on Pearson r and MAE as well. Pearson r is systematically higher than the square root of R^2 because r is insensitive to systematic offset and scale calibration—precisely the failure mode the main text identifies in the alloy-context contribution. Per-model numerical breakdowns are given in Tables S4 and S5.

Significance of the Pearson correlations. Per-fold Pearson tests on $n_{\text{test}} = 15$ molecules have limited power, so we report instead the pooled Pearson test: for each (model, setting) cell we concatenate the out-of-fold predictions from all 3 target alloys $\times$ 5 folds into a single list of $n = 225$ molecule-level (prediction, truth) pairs and compute one two-sided Pearson test on that pool. This is the standard `cross_val_predict` interpretation and uses the same molecule-level predictions that produced the mean-of-fold r in Table S4. Table S14 reports the pooled p -value. Both unfiltered TL settings give $p < 10^{-30}$ for every model, confirming that the close- and far-unfiltered correlations are not a fluke of the small per-fold sample. The Exact and filtered settings are also significant for every fitted model except ChemProp at Exact ($p = 0.90$, the only non-significant cell in the table). The discrepancy with that cell’s mean-of-fold $r = +0.18 \pm 0.07$ in Table S4 reflects fold-to-fold scale inconsistency rather than no per-fold signal: each ChemProp fold individually shows a weak positive ranking with truth, but the absolute predictions across folds do not share a common IE scale, so the concatenated pool integrates to a near-zero r . This is the same ranking-vs-calibration phenomenon documented for k NN-Tanimoto in the main text, here amplified by ChemProp training on only ~ 60 target-alloy samples without any transfer support. Gemini 3.1 Pro reaches $p < 10^{-22}$ in *every* setting, including Exact and the filtered transfer settings, consistent with its higher in-context Pearson r in those regimes (Sec. 3.6 of the main text).

Table S14: Pooled p -value of the Pearson correlation between predicted and experimental IE, computed on the $n=225$ out-of-fold molecule-level predictions per (model, setting) cell (3 target alloys $\times$ 5 folds $\times$ 15 test molecules; $n=224$ for Gemini after dropping one molecule with no valid prediction). Two-sided Pearson test.

	Exact (60)	Close filt.	Close unfilt.	Far filt.	Far unfilt.
RF	$\sim 10^{-6}$	$\sim 10^{-6}$	$\sim 10^{-61}$	$\sim 10^{-6}$	$\sim 10^{-57}$
GBR	$\sim 10^{-7}$	$\sim 10^{-5}$	$\sim 10^{-77}$	$\sim 10^{-6}$	$\sim 10^{-66}$
MLP	$\sim 10^{-5}$	0.005	$\sim 10^{-65}$	$\sim 10^{-5}$	$\sim 10^{-61}$
kNN-Tan	$\sim 10^{-5}$	0.003	$\sim 10^{-66}$	$\sim 10^{-4}$	$\sim 10^{-56}$
ChemProp	0.90	0.002	$\sim 10^{-37}$	$\sim 10^{-6}$	$\sim 10^{-33}$
Gemini 3.1 Pro	$\sim 10^{-22}$	$\sim 10^{-23}$	$\sim 10^{-38}$	$\sim 10^{-23}$	$\sim 10^{-50}$